RAJALAKSHMI ENGINEERING COLLEGE
(AnAutonomous Institution,Affiliated to AnnaUniversity, Chennai)
Departmentof ArtificialIntelligence | V SemesterB.Tech AIDS

AD23532 – Principles of Data Science

Ex2: Data Discovery and Preparation

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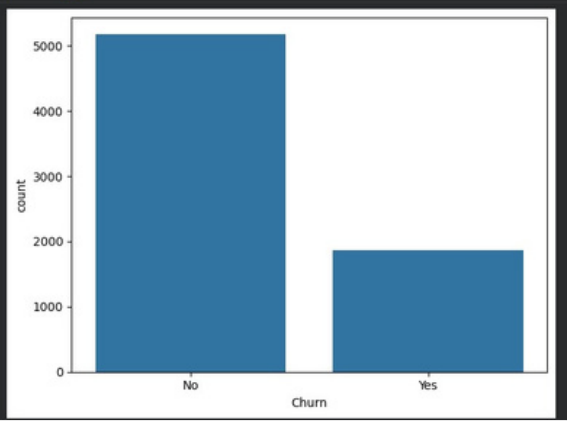
Task 1 – Reuse Ex:1 Steps 1–6, 10

Ex:1 Steps 1–6 and 10 (library import, data loading, exploration, cleaning, encoding, scaling and correlation heatmap) are reused
the Ex1 record.

Task 2 – Analyse the distribution of Customer Churn, Gender, Contract Type and Payment Method

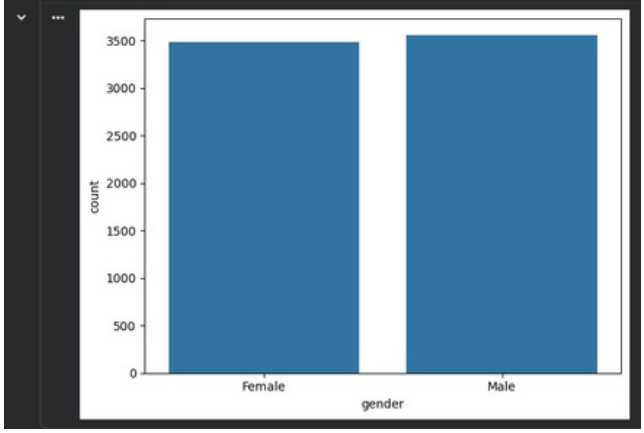
Customer Churn

```
sns.countplot(x="Churn",data=df)
plt.tight_layout()
plt.show()
```



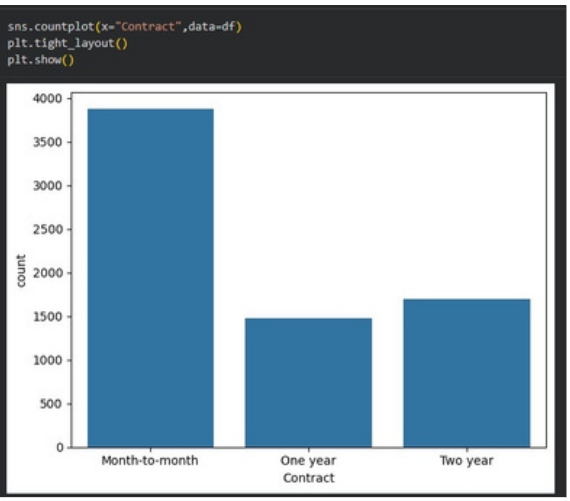
Gender

```
sns.countplot(x="gender",data=df)
plt.tight_layout()
plt.show()
```



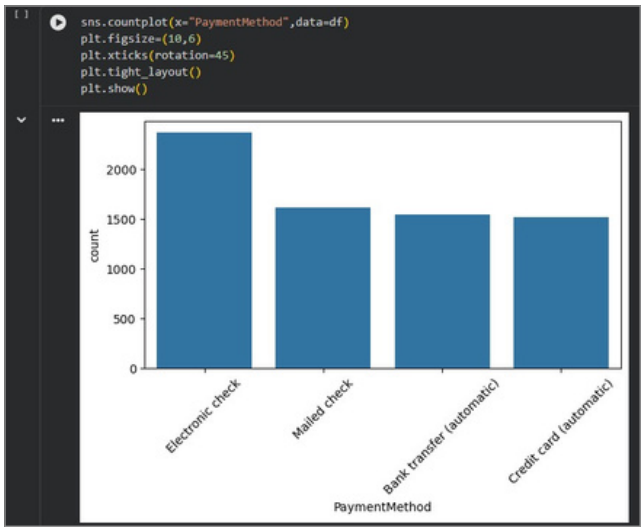
Contract Type

```
sns.countplot(x="Contract",data=df)
plt.tight_layout()
plt.show()
```



Payment Method

```
sns.countplot(x="PaymentMethod",data=df)
plt.figsize=(10,6)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Task 3 – Histograms and boxplots of MonthlyCharges, TotalCharges and Tenure

Code

```
# Visualize 'MonthlyCharges'
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1) # 1 row, 2 columns, 1st plot
plt.hist(df['MonthlyCharges'], bins=30, edgecolor='black')
plt.title('Distribution of Monthly Charges')
plt.xlabel('Monthly Charges')
plt.ylabel('Frequency')

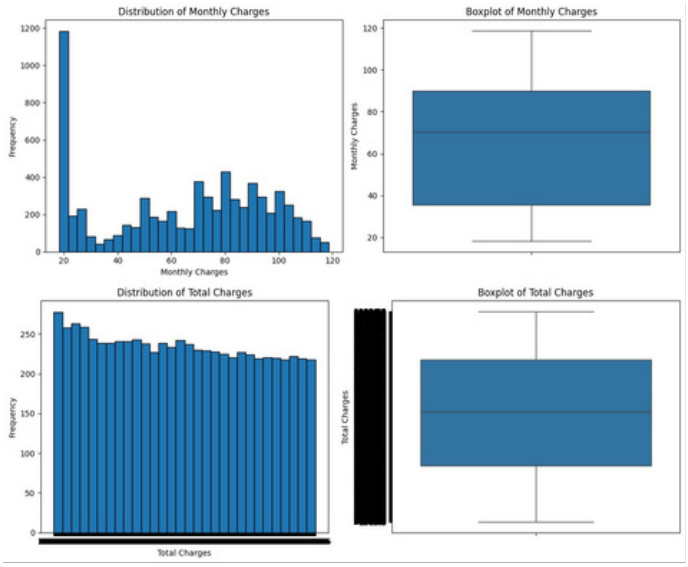
plt.subplot(1, 2, 2) # 1 row, 2 columns, 2nd plot
sns.boxplot(y=df['MonthlyCharges'])
plt.title('Boxplot of Monthly Charges')
plt.ylabel('Monthly Charges')
plt.tight_layout()
plt.show()

# Visualize 'TotalCharges'
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.hist(df['TotalCharges'], bins=30, edgecolor='black')
plt.title('Distribution of Total Charges')
plt.xlabel('Total Charges')
plt.ylabel('Frequency')

plt.subplot(1, 2, 2)
sns.boxplot(y=df['TotalCharges'])
plt.title('Boxplot of Total Charges')
plt.ylabel('Total Charges')
plt.tight_layout()
plt.show()

# Visualize 'tenure'
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.hist(df['tenure'], bins=30, edgecolor='black')
plt.title('Distribution of Tenure')
plt.xlabel('Tenure (months)')
plt.ylabel('Frequency')
```

Output (MonthlyCharges,TotalCharges)



Task 4 – Detect outliers using the Interquartile Range (IQR) method

Code

4. Detect outliers in numerical features using the Interquartile Range (IQR) method.

```
numerical_cols = df.select_dtypes(include=np.number).columns

outlier_summary = {}

for col in numerical_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    num_outliers = len(outliers)

    outlier_summary[col] = {
        'Q1': Q1,
        'Q3': Q3,
        'IQR': IQR,
        'Lower Bound': lower_bound,
        'Upper Bound': upper_bound,
        'Number of Outliers': num_outliers,
        'Percentage of Outliers': (num_outliers / len(df)) * 100
    }

    print(f"\n--- Outlier Analysis for {col} ---")
    print(f"Q1: {Q1:.2f}")
    print(f"Q3: {Q3:.2f}")
    print(f"IQR: {IQR:.2f}")
    print(f"Lower Bound: {lower_bound:.2f}")
    print(f"Upper Bound: {upper_bound:.2f}")
```

Output – outlier summary

```
outlier_df = pd.DataFrame(outlier_summary).T
display(outlier_df)
```

```
--- Outlier Analysis for SeniorCitizen ---
Q1: 0.00
Q3: 0.00
IQR: 0.00
Lower Bound: 0.00
Upper Bound: 0.00
Number of Outliers: 1142
Percentage of Outliers: 16.21%

--- Outlier Analysis for tenure ---
Q1: 9.00
Q3: 55.00
IQR: 46.00
Lower Bound: -60.00
Upper Bound: 124.00
Number of Outliers: 0
Percentage of Outliers: 0.00%

--- Outlier Analysis for MonthlyCharges ---
Q1: 35.50
Q3: 89.85
IQR: 54.35
Lower Bound: -46.02
Upper Bound: 171.38
Number of Outliers: 0
Percentage of Outliers: 0.00%
```

	Q1	Q3	IQR	Lower Bound	Upper Bound	Number of Outliers	Percentage of Outliers
SeniorCitizen	0.0	0.00	0.00	0.000	0.000	1142.0	16.214681
tenure	9.0	55.00	46.00	-60.000	124.000	0.0	0.000000
MonthlyCharges	35.5	89.85	54.35	-46.025	171.375	0.0	0.000000

Task 6 – Analytical questions

Q1.Whichcontracttypehasthehighestcustomer churn? Month-to-month (churn rate ≈ 42.7%).

Which contract type has the highest customer churn?

```
churn_by_contract = df.groupby(['Contract', 'Churn']).size().unstack(fill_value=0)
churn_by_contract['Churn_Rate'] = churn_by_contract['Yes'] / (churn_by_contract['No'] + churn_by_contract['Yes'])
display(churn_by_contract)

highest_churn_contract = churn_by_contract['Churn_Rate'].idxmax()
print(f"The contract type with the highest churn rate is: {highest_churn_contract}")
```

Contract	No	Yes	Churn_Rate
Month-to-month	2220	1655	0.427097
One year	1307	166	0.112695
Two year	1647	48	0.028319

```
The contract type with the highest churn rate is: Month-to-month
```

Q2. Which payment method is most frequently used by customers who churn? Electronic check (1071 churned customers).

Which payment method is most frequently used by customers who churn?

```
churned_customers = df[df['Churn'] == 'Yes']
most_frequent_payment_method = churned_customers['PaymentMethod'].value_counts().idxmax()
print(f"The payment method most frequently used by customers who churn is: {most_frequent_payment_method}")
display(churned_customers['PaymentMethod'].value_counts())
```

```
The payment method most frequently used by customers who churn is: Electronic check
```

PaymentMethod	count
Electronic check	1071
Mailed check	308
Bank transfer (automatic)	258
Credit card (automatic)	232

dtype: int64

Q3. Do customers with shorter tenure tend to churn more often? Yes – average tenure is 17.98 months for churned vs 37.57 months for non-churned customers.

Do customers with shorter tenure tend to churn more often?

```
mean_tenure_churn_yes = df[df['Churn'] == 'Yes']['tenure'].mean()
mean_tenure_churn_no = df[df['Churn'] == 'No']['tenure'].mean()

print(f"Average tenure for churned customers: {mean_tenure_churn_yes:.2f} months")
print(f"Average tenure for non-churned customers: {mean_tenure_churn_no:.2f} months")

if mean_tenure_churn_yes < mean_tenure_churn_no:
    print("Yes, customers with shorter tenure tend to churn more often.")
else:
    print("No, customers with shorter tenure do not tend to churn more often (or the difference is negligible).")
```

```
Average tenure for churned customers: 17.98 months
Average tenure for non-churned customers: 37.57 months
Yes, customers with shorter tenure tend to churn more often.
```

Task 7 – Split the cleaned dataset into Training (80%) and Testing (20%)

Train/test split using `train_test_split(X, y, test_size=0.2, random_state=42)`

```
from sklearn.model_selection import train_test_split

# Define features (X) and target (y)
X = df_encoded.drop(['Churn_Yes', 'customerID'], axis=1) # Drop 'customerID' as it's not a feature
y = df_encoded['Churn_Yes']

# Split the dataset into training (80%) and testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Training set shape (X_train, y_train):")
```

Task 8 – Save the final processed dataset

Processed dataset saved as `CustomerChurn_Preprocessed.csv`

```
# Save the final processed dataset as a new CSV file
df_encoded.to_csv('CustomerChurn_Preprocessed.csv', index=False)
print("Processed dataset saved to 'CustomerChurn_Preprocessed.csv'")
```

```
Processed dataset saved to 'CustomerChurn_Preprocessed.csv'
```